
React & Modern JavaScript

Build powerful single-page applications with React, hooks, and the modern JS ecosystem.

Level: Intermediate | Estimated time: 12 hours | Category: Development

Course overview

Build powerful single-page applications with React, hooks, and the modern JS ecosystem.

This PDF course pack was generated for the project from publicly available internet resources and the app's built-in lesson outline.

Lesson 1: React Fundamentals

React is a JavaScript library for building user interfaces.

- JSX syntax and Babel transpilation
- Creating and nesting components
- Props: passing data down to child components
- State with useState hook
- Rendering lists with .map() and key prop
- Handling events in React
- Conditional rendering
- Component lifecycle and useEffect
- Basic form handling with controlled inputs

Key concept: React uses a virtual DOM to efficiently update only the parts of the UI that change.

Lesson 2: Hooks & State Management

React hooks make it easy to add state and side effects to function components.

- useState for local component state
- useEffect for side effects (data fetching, subscriptions)
- useContext for sharing state without prop drilling
- useReducer for complex state logic
- Custom hooks: extracting reusable logic
- React Context API for global state
- Introduction to Zustand / Redux Toolkit

Key concept: Hooks let you use state and React features in functional components without writing a class.

Further reading from the internet

These official or widely used references match the course topic and can be used for deeper study.

React: Describing the UI

<https://react.dev/learn/describing-the-ui>

React: State - a component's memory

<https://react.dev/learn/state-a-components-memory>

React: Managing state

<https://react.dev/learn/managing-state>